

CHAPTER 1 COMMANDS

```
preethi_r140306@cloudshell:~$ uname
Linux
preethi_r140306@cloudshell:~$ uptime
10:12:09 up 3:19, 0 user, load average: 0.02, 0.10, 0.06
preethi_r140306@cloudshell:~$ hostname
cs-720557602863-default
preethi_r140306@cloudshell:~$ dmesg
dmesg: read kernel buffer failed: Operation not permitted
preethi_r140306@cloudshell:~$ free
              total        used        free      shared  buff/cache   available
Mem:      16382844      2296144      12490600          1136       1877072      14086700
Swap:              0              0
preethi_r140306@cloudshell:~$ top
top - 10:13:17 up 3:20, 0 user, load average: 0.00, 0.08, 0.05
Tasks: 35 total, 1 running, 34 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.2 us, 0.3 sy, 0.0 ni, 99.3 id, 0.0 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 15998.9 total, 12193.0 free, 2247.0 used, 1833.3 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 13751.9 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
482	root	20	0	1867212	46036	32264	S	0.7	0.3	0:05.72	containerd
1549	preethi+	20	0	1136912	81200	48464	S	0.7	0.5	0:05.79	node
439	root	20	0	1234524	12544	8336	S	0.3	0.1	0:01.11	editor-proxy
936	preethi+	20	0	11.3g	121324	51220	S	0.3	0.7	0:08.49	node
1	root	20	0	4324	3268	2988	S	0.0	0.0	0:00.07	bash
9	syslog	20	0	222140	3808	2440	S	0.0	0.0	0:00.04	rsyslogd
25	root	20	0	38768	28088	10980	S	0.0	0.2	0:01.10	python
26	root	20	0	6196	2668	2428	S	0.0	0.0	0:00.00	logger
67	root	10	-10	12024	4156	3040	S	0.0	0.0	0:00.04	sshd
353	root	20	0	2161448	85284	61744	S	0.0	0.5	0:00.89	dockerd
440	root	20	0	16328	6324	5400	S	0.0	0.0	0:00.01	sudo
443	root	20	0	1226356	2820	1976	S	0.0	0.0	0:00.00	tmux-agent
692	root	20	0	2696	1528	1432	S	0.0	0.0	0:00.00	sleep

\$top

```
0[|
1[|
2[|
3[|
Mem[|||||||
Swap[|||||||
0.7% Tasks: 35, 110 thr, 0 kthr; 1 running
0.7% Load average: 0.00 0.03 0.03
0.7% Uptime: 03:24:27
0.7%
1.91G/15.6G
0K/0K
```

Main	I/O										
PID	USER	PRI	NI	VIRT	RES	SHR	S	CPU%	MEM%	TIME+	Command
1	root	20	0	4324	3268	2988	S	0.0	0.0	0:00.05	/bin/bash /google/scripts/onrun.sh sleep infinity
9	syslog	20	0	216M	3808	2440	S	0.0	0.0	0:00.00	/usr/sbin/rsyslogd
10	syslog	20	0	216M	3808	2440	S	0.0	0.0	0:00.02	/usr/sbin/rsyslogd
11	syslog	20	0	216M	3808	2440	S	0.0	0.0	0:00.00	/usr/sbin/rsyslogd
12	syslog	20	0	216M	3808	2440	S	0.0	0.0	0:00.01	/usr/sbin/rsyslogd
25	root	20	0	38768	28088	10980	S	0.0	0.2	0:01.14	/usr/bin/python /usr/bin/supervisord -n -c /google/devshell/supervisord.conf --pidfile=/var/run/supervisor.pid --logfile=/var/log/supe
26	root	20	0	6196	2668	2428	S	0.0	0.0	0:00.00	logger -t supervisord
67	root	10	-10	12024	4156	3040	S	0.0	0.0	0:00.04	sshd: /usr/sbin/sshd -p 22 -o AuthorizedKeysFile=/etc/ssh/keys/authorized_keys [listener] 0 of 10-60 startups
353	root	20	0	2110M	85284	61744	S	0.0	0.5	0:00.19	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
410	root	20	0	2110M	85284	61744	S	0.0	0.5	0:00.13	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
411	root	20	0	2110M	85284	61744	S	0.0	0.5	0:00.10	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
412	root	20	0	2110M	85284	61744	S	0.0	0.5	0:00.00	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
413	root	20	0	2110M	85284	61744	S	0.0	0.5	0:00.00	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
414	root	20	0	2110M	85284	61744	S	0.0	0.5	0:00.00	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
437	root	20	0	2110M	85284	61744	S	0.0	0.5	0:00.05	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
439	root	20	0	1205M	12544	8336	S	0.0	0.1	0:00.12	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content -etag-f
440	root	20	0	16328	6324	5400	S	0.0	0.0	0:00.01	sudo /google/devshell/tmux-agent
442	root	20	0	1205M	12544	8336	S	0.0	0.1	0:00.31	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content -etag-f
443	root	20	0	1197M	2820	1976	S	0.0	0.0	0:00.00	/google/devshell/tmux-agent
444	root	20	0	1205M	12544	8336	S	0.0	0.1	0:00.00	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content -etag-f
445	root	20	0	1205M	12544	8336	S	0.0	0.1	0:00.00	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content -etag-f
446	root	20	0	1205M	12544	8336	S	0.0	0.1	0:00.15	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content -etag-f

```
preethi_r140306@cloudshell:~$ lsb_release
No LSB modules are available.
preethi_r140306@cloudshell:~$ arch
x86_64
```

CHAPTER 2 COMMANDS

```
preethi_r140306@cloudshell:~$ mkdir preethi
preethi_r140306@cloudshell:~$ mkdir pree
preethi_r140306@cloudshell:~$ pwd
/home/preethi_r140306
preethi_r140306@cloudshell:~$ ls
nano.2505.save  pree  preethi  README-cloudshell.txt
preethi_r140306@cloudshell:~$ cd preethi
preethi_r140306@cloudshell:~/preethi$ cd
preethi_r140306@cloudshell:~$ rmdir pree
preethi_r140306@cloudshell:~$ touch file1.txt
preethi_r140306@cloudshell:~$ echo "Hello" > file1.txt
preethi_r140306@cloudshell:~$ cp file1.txt copy.txt
preethi_r140306@cloudshell:~$ rm copy.txt
preethi_r140306@cloudshell:~$ mv file1.txt file.txt
preethi_r140306@cloudshell:~$ ln file.txt file2.txt
preethi_r140306@cloudshell:~$ cat file2.txt
Hello
preethi_r140306@cloudshell:~$ rm file2.txt
preethi_r140306@cloudshell:~$ cat file.txt
Hello
```

CHAPTER 3 COMMANDS

```
preethi_r140306@cloudshell:~$ echo "Hello" > file1.txt
preethi_r140306@cloudshell:~$ echo "Everyone" > file2.txt
preethi_r140306@cloudshell:~$ cat file1.txt file2.txt > combined.txt
preethi_r140306@cloudshell:~$ cat combined.txt
Hello
Everyone
preethi_r140306@cloudshell:~$ echo "Earth is the third planet from the Sun and the only known place in the universe to harbor life, serving as our shared home. Covered
iverse atmosphere, vibrant ecosystems, and rocky terrain that supports millions of species. As a dynamic planet constantly evolving through tectonic activity, it offer
which we must protect from threats like pollution." > large.txt
preethi_r140306@cloudshell:~$ more large.txt
Earth is the third planet from the Sun and the only known place in the universe to harbor life, se
rving as our shared home. Covered largely by water, it features a diverse atmosphere, vibrant ecos
ystems, and rocky terrain that supports millions of species. As a dynamic planet constantly evolvi
ng through tectonic activity, it offers the perfect conditions for life, which we must protect fro
m threats like pollution.
```

\$less large.txt (To check the pattern : ?a)

Earth is the third planet from the Sun and the only known place in the universe to harbor life, serving as our shared home. Covered largely by water, it supports millions of species. As a dynamic planet constantly evolving through tectonic activity, it offers the perfect conditions for life.

~ ~ ~ ~ ~

Earth is the third planet from the Sun and the only known place in the universe to harbor life, serving as our shared home. Covering a vast, diverse, and rocky terrain that supports millions of species. As a dynamic planet constantly evolving through tectonic activity, it also faces challenges like pollution.

```
preethi r140306@cloudshell:~$ less large.txt
```

```
preethi r140306@cloudshell:~$ head large.txt
```

Earth is the third planet from the Sun and the only known place in the universe to harbor life, serving as our shared home. Covered largely by water, it features a diverse atmosphere, s, and rocky terrain that supports millions of species. As a dynamic planet constantly evolving through tectonic activity, it offers the perfect conditions for life, which we must protect from pollution.

```
preethi r140306@cloudshell:~$ tail large.txt
```

Earth is the third planet from the Sun and the only known place in the universe to harbor life, serving as our shared home. Covered largely by water, it features a diverse atmosphere, s, and rocky terrain that supports millions of species. As a dynamic planet constantly evolving through tectonic activity, it offers the perfect conditions for life, which we must protect from pollution.

\$nano large.txt

```
preethi r140306@cloudshell:~$ cat large.txt
```

Earth is the third planet from the Sun and the only known place in the universe to harbor life, serving as our shared home. Covered largely by water, it features a s, and rocky terrain that supports millions of species. As a dynamic planet constantly evolving through tectonic activity, it offers the perfect conditions for lif like pollution.

"Save Earth Save Life!".

```
preethi r140306@cloudshell:~$ vim large.txt
```

```
preethi r140306@cloudshell:~$ cat large.txt
```

Earth is the third planet from the Sun and the only known place in the universe to harbor life, serving as our shared home. <"Data is inserted by using vim"> Covered largely by water, dense atmosphere, vibrant ecosystems, and rocky terrain that supports millions of species. As a dynamic planet constantly evolving through tectonic activity, it offers the perfect conditions which we must protect from threats like pollution.

"Save Earth Save Life!".

```
preethi r140306@cloudshell:~$ gedit large.txt
```

```
bash: gedit: command not found
```

```
preethi_r140306@cloudshell:~$ awk '{print $1}' large.txt
```

Earth

"Save

```
preethi r140306@cloudshell:~$ awk -F '{print $1}' large.txt
```

```
awk: 1: unexpected character '.'
```

```
preethi_r140306@cloudshell:~$ sed 's/Earth/The Earth/' large.txt
The Earth is the third planet from the Sun and the only known place in the universe to harbor life, serving as our shared home. <"Data is inserted by using vim"> Covered largely by water,
a diverse atmosphere, vibrant ecosystems, and rocky terrain that supports millions of species. As a dynamic planet constantly evolving through tectonic activity, it offers the perfect cond
fe, which we must protect from threats like pollution.
"Save The Earth Save Life!".

preethi_r140306@cloudshell:~$ sed 's/a/^a/g' large.txt
E^arth is the third pl^anet from the Sun ^and the only known pl^ace in the universe to h^arbor life, serving ^as our sh^ared home. <"D^at^a is inserted by using vim"> Covered l^argely by
^atures ^a diverse ^atmosphere, vibr^ant ecosystems, ^and rocky terr^ain th^at supports millions of species. As ^a dyn^amic pl^anet const^antly evolving through tectonic ^activity, it offe
t conditions for life, which we must protect from thre^ats like pollution.
"S^ave E^arth S^ave Life!".
```

CHAPTER 4 COMMANDS

```
preethi_r140306@cloudshell:~$ touch demo.txt
preethi_r140306@cloudshell:~$ ls -l demo.txt
-rw-r--r-- 1 preethi_r140306 preethi_r140306 0 May 14 17:31 demo.txt
preethi_r140306@cloudshell:~$ chmod +x demo.txt
preethi_r140306@cloudshell:~$ ls -l demo.txt
-rwxr-xr-x 1 preethi_r140306 preethi_r140306 0 May 14 17:31 demo.txt
preethi_r140306@cloudshell:~$ chmod 755 demo.txt
preethi_r140306@cloudshell:~$ ls -l demo.txt
-rwxr-xr-x 1 preethi_r140306 preethi_r140306 0 May 14 17:31 demo.txt
preethi_r140306@cloudshell:~$ sudo chown $USER demo.txt
preethi_r140306@cloudshell:~$ sudo chgrp $USER demo.txt
preethi_r140306@cloudshell:~$ ls -l demo.txt
-rwxr-xr-x 1 preethi_r140306 preethi_r140306 0 May 14 17:31 demo.txt
preethi_r140306@cloudshell:~$ umask
0022
preethi_r140306@cloudshell:~$ touch test.txt
preethi_r140306@cloudshell:~$ ls -l test.txt
-rw-r--r-- 1 preethi_r140306 preethi_r140306 0 May 14 17:39 test.txt
preethi_r140306@cloudshell:~$ stat demo.txt
  File: demo.txt
  Size: 0          Blocks: 0          IO Block: 4096   regular empty file
Device: 8,17      Inode: 131491      Links: 1
Access: (0755/-rwxr-xr-x)  Uid: ( 1000/preethi_r140306)   Gid: ( 1000/preethi_r140306)
Access: 2026-05-14 17:31:06.491638642 +0000
Modify: 2026-05-14 17:31:06.491638642 +0000
Change: 2026-05-14 17:38:07.878050784 +0000
Birth: 2026-05-14 17:31:06.491638642 +0000

preethi_r140306@cloudshell:~$ getfacl demo.txt
bash: getfacl: command not found
iemini Code Assist Outline dshell:~$ setfacl -m u:$USER:rwx demo.txt
basn: setrac1: command not found
preethi_r140306@cloudshell:~$ lsattr demo.txt
-----e----- demo.txt
preethi_r140306@cloudshell:~$ sudo chattr +i demo.txt
preethi_r140306@cloudshell:~$ rm demo.txt
rm: cannot remove 'demo.txt': Operation not permitted
preethi_r140306@cloudshell:~$ sudo chattr -i demo.txt
preethi_r140306@cloudshell:~$ rm demo.txt
```

CHAPTER 5 COMMANDS

```
preethi_r140306@cloudshell:~$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1460
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 1a:71:9e:88:c7:16 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1460
    inet 10.88.0.3 netmask 255.255.0.0 broadcast 10.88.255.255
    ether 96:1a:91:f2:aa:d6 txqueuelen 1000 (Ethernet)
    RX packets 3762 bytes 3554002 (3.5 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 3705 bytes 17351270 (17.3 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 5098 bytes 38970019 (38.9 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 5098 bytes 38970019 (38.9 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
preethi_r140306@cloudshell:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: eth0@if7: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1460 qdisc noqueue state UP group default qlen 1000
    link/ether 96:1a:91:f2:aa:d6 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.88.0.3/16 brd 10.88.255.255 scope global eth0
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1460 qdisc noqueue state DOWN group default
    link/ether 1a:71:9e:88:c7:16 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
preethi_r140306@cloudshell:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=0.493 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=114 time=0.381 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=114 time=0.301 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=114 time=0.344 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3067ms
rtt min/avg/max/mdev = 0.301/0.379/0.493/0.071 ms
preethi_r140306@cloudshell:~$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:59846         localhost:ssh           ESTABLISHED
```

tcp	0	0	localhost:59846	localhost:ssh	ESTABLISHED
tcp	0	0	localhost:971	localhost:60248	ESTABLISHED
tcp	0	0	localhost:971	localhost:60232	ESTABLISHED
tcp	0	0	localhost:58100	localhost:981	TIME_WAIT
tcp	0	0	localhost:56208	localhost:981	TIME_WAIT
tcp	0	0	localhost:54668	localhost:981	TIME_WAIT
tcp	0	0	localhost:59842	localhost:ssh	ESTABLISHED
tcp	0	0	localhost:60248	localhost:971	ESTABLISHED
tcp	0	0	localhost:60232	localhost:971	ESTABLISHED
tcp	0	0	localhost:51936	localhost:970	ESTABLISHED
tcp	0	0	cs-720557602863-d:55630	sa-in-f101.1e100.:https	ESTABLISHED
tcp	0	0	localhost:38124	localhost:ssh	TIME_WAIT
tcp	0	0	localhost:ssh	localhost:59842	ESTABLISHED
tcp	0	0	localhost:51944	localhost:970	ESTABLISHED
tcp	0	0	localhost:ssh	localhost:59846	ESTABLISHED
tcp6	0	0	cs-720557602863-def:980	10.180.213.13:42132	ESTABLISHED
tcp6	0	0	cs-720557602863-def:980	10.180.213.13:55654	ESTABLISHED
tcp6	0	0	localhost:970	localhost:51944	ESTABLISHED
tcp6	0	0	cs-720557602863-def:980	10.180.212.13:51816	ESTABLISHED
tcp6	0	0	cs-720557602863-def:980	10.180.213.13:55640	ESTABLISHED
tcp6	0	0	localhost:970	localhost:51936	ESTABLISHED
tcp6	0	0	cs-720557602863-def:980	10.180.212.13:51798	ESTABLISHED
tcp6	0	0	cs-720557602863-def:980	10.180.212.13:51808	ESTABLISHED

Active UNIX domain sockets (w/o servers)

Proto	RefCnt	Flags	Type	State	I-Node	Path
unix	2	[]	DGRAM	CONNECTED	77251	
unix	3	[]	STREAM	CONNECTED	79873	
unix	3	[]	STREAM	CONNECTED	77926	/var/run/docker/containerd/containerd.sock
unix	3	[]	STREAM	CONNECTED	/9900	
unix	3	[]	STREAM	CONNECTED	76746	
unix	3	[]	STREAM	CONNECTED	77734	
unix	3	[]	STREAM	CONNECTED	77737	
unix	3	[]	STREAM	CONNECTED	80249	
unix	3	[]	STREAM	CONNECTED	78000	
unix	3	[]	STREAM	CONNECTED	77733	
unix	3	[]	STREAM	CONNECTED	76800	
unix	2	[]	DGRAM	CONNECTED	76284	
unix	3	[]	STREAM	CONNECTED	78442	
unix	3	[]	STREAM	CONNECTED	80252	
unix	9	[]	DGRAM	CONNECTED	77003	/dev/log
unix	3	[]	STREAM	CONNECTED	77735	
unix	3	[]	STREAM	CONNECTED	78443	
unix	3	[]	STREAM	CONNECTED	80246	
unix	2	[]	STREAM	CONNECTED	78898	
unix	3	[]	STREAM	CONNECTED	78857	/var/run/docker/containerd/containerd.sock
unix	3	[]	STREAM	CONNECTED	77782	
unix	3	[]	STREAM	CONNECTED	76544	
unix	3	[]	STREAM	CONNECTED	80250	
unix	3	[]	STREAM	CONNECTED	76539	/var/run/docker/containerd/containerd.sock
unix	3	[]	STREAM	CONNECTED	80247	
unix	2	[]	DGRAM	CONNECTED	77995	
unix	3	[]	STREAM	CONNECTED	79901	
unix	3	[]	STREAM	CONNECTED	77740	

CHAPTER 6

```
preethi_r140306@cloudshell:~$ mkdir chapter^_process
preethi_r140306@cloudshell:~$ ps
  PID TTY          TIME CMD
 2108 pts/3        00:00:00 bash
 2550 pts/3        00:00:00 ps
preethi_r140306@cloudshell:~$ pgrep bash
1
720
723
750
755
2108
preethi_r140306@cloudshell:~$ sleep 300
^C
preethi_r140306@cloudshell:~$ sleep 300 &
[1] 2642
preethi_r140306@cloudshell:~$ jobs
[1]+  Running                  sleep 300 &
preethi_r140306@cloudshell:~$ fg
sleep 300
^C
preethi_r140306@cloudshell:~$ bg
bash: bg: current: no such job
preethi_r140306@cloudshell:~$ pkill sleep
pkill: killing pid 685 failed: Operation not permitted
preethi_r140306@cloudshell:~$ nice -n 10 sleep 200 &
[1] 2868
```

```
preethi_r140306@cloudshell:~$ nice -n 10 sleep 200 &
[1] 2868
preethi_r140306@cloudshell:~$ pgrep sleep
685
2868
preethi_r140306@cloudshell:~$ renice 5 4513
renice: failed to get priority for 4513 (process ID): No such process
preethi_r140306@cloudshell:~$ █
```

CHAPTER 7

Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT\_ID]`.
You can view your projects by running `gcloud projects list`.

```
preethi_r140306@cloudshell:~$ mkdir chapter7
```

```
preethi_r140306@cloudshell:~$ df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
overlay	95G	73G	22G	78%	/
tmpfs	64M	0	64M	0%	/dev
/dev/sda1	95G	73G	22G	78%	/root
/dev/disk/by-id/google-home-part1	4.8G	155M	4.4G	4%	/home
/dev/root	2.0G	1.3G	715M	64%	/usr/lib/modules
shm	64M	0	64M	0%	/dev/shm

```
preethi_r140306@cloudshell:~$ lsblk
```

```
bash: lsblk: command not found
```

```
preethi_r140306@cloudshell:~$ sudo fdisk -l
```

```
sudo: fdisk: command not found
```

```
preethi_r140306@cloudshell:~$ sudo tune2fs -l /dev/sda1
```

```
tune2fs 1.47.0 (5-Feb-2023)
```

```
tune2fs: invalid option -- 'l'
```

```
Usage: tune2fs [-c max_mounts_count] [-e errors_behavior] [-f] [-g group]
        [-i interval[d|m|w]] [-j] [-J journal_options] [-l]
        [-m reserved_blocks_percent] [-o [^]mount_options[,...]]
        [-r reserved_blocks_count] [-u user] [-C mount_count]
        [-L volume_label] [-M last_mounted_dir]
        [-O [^]feature[,...]] [-Q quota_options]
        [-E extended-option[,...]] [-T last_check_time] [-U UUID]
        [-I new_inode_size] [-z undo_file] device
```

```
preethi_r140306@cloudshell:~$
```

CHAPTER 8

\$ Sudo apt install sysstat dstat glances y

```
Get:90 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-cssselect all 1.2.0-2 [18.5 kB]
Fetched 61.4 MB in 2s (37.9 MB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package libsigsegv2:amd64.
(Reading database ... 121190 files and directories currently installed.)
Preparing to unpack .../libsigsegv2_2.14-1ubuntu2_amd64.deb ...
Unpacking libsigsegv2:amd64 (2.14-1ubuntu2) ...
Setting up libsigsegv2:amd64 (2.14-1ubuntu2) ...
Selecting previously unselected package gawk.
(Reading database ... 121197 files and directories currently installed.)
Preparing to unpack .../00-gawk_1%3a5.2.1-2build3_amd64.deb ...
Unpacking gawk (1:5.2.1-2build3) ...
Selecting previously unselected package python3-defusedxml.
Preparing to unpack .../01-python3-defusedxml_0.7.1-2_all.deb ...
Unpacking python3-defusedxml (0.7.1-2) ...
Selecting previously unselected package python3-psutil.
Preparing to unpack .../02-python3-psutil_5.9.8-2build2_amd64.deb ...
Unpacking python3-psutil (5.9.8-2build2) ...
Selecting previously unselected package libdouble-conversion3:amd64.
Preparing to unpack .../03-libdouble-conversion3_3.3.0-1build1_amd64.deb ...
Unpacking libdouble-conversion3:amd64 (3.3.0-1build1) ...
Selecting previously unselected package python3-urllib3:amd64.
Preparing to unpack .../04-python3-urllib3_5.9.0-1build1_amd64.deb ...
Unpacking python3-urllib3:amd64 (5.9.0-1build1) ...
Selecting previously unselected package glances.
Preparing to unpack .../05-glances_3.4.0.3+dfsg-1_all.deb ...
Unpacking glances (3.4.0.3+dfsg-1) ...
Selecting previously unselected package python3-netifaces:amd64.
```

```
preethi_r140306@cloudshell:~$ iostat
```

```
Linux 6.6.137+ (cs-720557602863-default) 05/19/2026 _x86_64_ (4 CPU)
```

avg-cpu:	%user	%nice	%system	%iowait	%steal	%idle
	0.71	0.00	0.40	0.07	0.02	98.79

Device	tps	kB_read/s	kB_wrtn/s	kB_dscd/s	kB_read	kB_wrtn	kB_dscd
dm-0	0.84	46.01	0.00	0.00	435476	0	0
sda	5.47	179.85	86.15	0.00	1702111	815289	0
sdb	0.11	13.97	2.38	0.00	132233	22520	0

CHAPTER 9

Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT\_ID]`.
You can view your projects by running `gcloud projects list`.

```
preethi_r140306@cloudshell:~$ sudo useradd testuser
preethi_r140306@cloudshell:~$ sudo usermod -aG sudo testuser
preethi_r140306@cloudshell:~$ groups
preethi_r140306 adm sudo docker
preethi_r140306@cloudshell:~$ sudo groupadd developers
preethi_r140306@cloudshell:~$ sudo usermod -aG developers testuser
preethi_r140306@cloudshell:~$ su testuser
```

Password:

su: Authentication failure

```
preethi_r140306@cloudshell:~$ iostat
```

Linux 6.6.137+ (cs-720557602863-default) 05/19/2026 _x86_64_ (4 CPU)

avg-cpu:	%user	%nice	%system	%iowait	%steal	%idle
	0.73	0.00	0.42	0.07	0.02	98.76

Device	tps	kB_read/s	kB_wrtn/s	kB_dscd/s	kB_read	kB_wrtn	kB_dscd
dm-0	0.80	43.45	0.00	0.00	435476	0	0
sda	5.44	169.86	88.17	0.00	1702275	883573	0
sdb	0.11	13.19	2.28	0.00	132233	22852	0

```
preethi_r140306@cloudshell:~$ █
```

CHAPTER 10

```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
preethi_r140306@cloudshell:~$ mkdir chapter10
preethi_r140306@cloudshell:~$ cd chapter10
preethi_r140306@cloudshell:~/chapter10$ touch data.txt
preethi_r140306@cloudshell:~/chapter10$ bzip2 data.txt
preethi_r140306@cloudshell:~/chapter10$ xz data.txt
xz: data.txt: No such file or directory
preethi_r140306@cloudshell:~/chapter10$ tar -cvf archive.tar data.txt
tar: data.txt: Cannot stat: No such file or directory
tar: Exiting with failure status due to previous errors
preethi_r140306@cloudshell:~/chapter10$ zip archive.zip data.txt
  zip warning: name not matched: data.txt

zip error: Nothing to do! (archive.zip)
preethi_r140306@cloudshell:~/chapter10$
```